

removing water. If we don't fill it up all the way or if the water spills out of the cup, the level of the water in the sigma bucket changes. The result limits the precision of the sense. One other limiting factor is the sigma bucket. If it is "leaky" and the water it holds leaks out, the quality of the sense will be affected.

Table 17.1 An example of DSM.

Time, seconds	Water level in the sigma bucket (cups)	Remove cup? (water level > 5?)	Running average
0	5		0
10	5.25	Yes	1
20	4.5	No	0.5
30	4.75	No	0.33
40	5	No	0.25
50	5.25	Yes	0.4
60	4.5	No	0.33
70	4.75	No	0.29
80	5	No	0.25
90	5.25	Yes	0.33
100	4.5	No	0.3
110	4.75	No	0.27
120	5	No	0.25
130	5.25	Yes	0.31
140	4.5	No	0.29
150	4.75	No	0.26
160	5	No	0.25
170	5.25	Yes	0.29

The Counter

We might wonder how, in a practical sensing circuit, we can get decimal numbers like the ones seen in Table 17.1? The answer to this question is seen in Fig. 17.3. If a "yes, remove a cup of water" is indicated by the output of the DSM sensing circuit going high, a counter can be used to generate the number. If N is the total number of times we clock the DSM, then

$$\text{Output number} = \frac{\text{number of Yes outputs}}{N} \quad (17.1)$$

In Table 17.1 N is 17. The number of "yes" outputs is 5. The final output number is then 5/17 or 0.29. If a 10-bit counter is used (assuming it was reset to zeroes at the beginning of the sense), then its digital output is 00 0000 0101. To find the absolute value of the input, we multiply this number by the size of the feedback signal (here one cup) to get

$$\text{Input signal} = \text{Output number} \times \text{fed back signal size} \quad (17.2)$$

or here 0.29 cups (per 10 seconds). Note that to converge on the actual rate of 0.25 cups/10 seconds we would need to average more points.

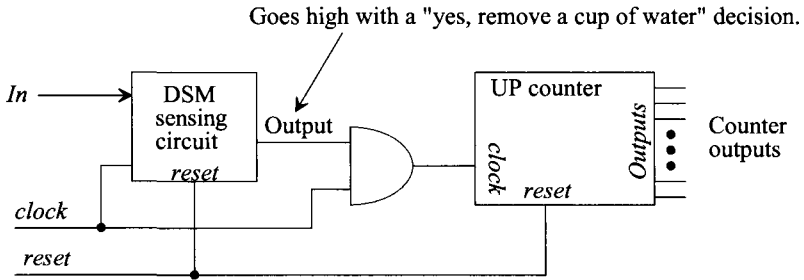


Figure 17.3 How a counter is used to average the outputs of a DSM sensing circuit.

Cup Size

The size of our fed back signal, here the cup size, is very important. If we use a small cup, we can converge on the correct digital representation (the counter output) of our analog signal (the flow of water into the sigma bucket) quicker. However, if we use too small of a cup, then we can't remove the water fast enough from the bucket and it will overflow. For a large range, we need a big cup. Using a big cup means that we have to average longer (the sensing lasts for a longer period of time).

Another Example

Figure 17.4 shows another water analogy where DSM can be used for measuring an analog quantity (in this case the water flowing out of the sigma bucket). When the water level gets too low, we add a cup of water to the bucket. Again, averaging how often we add the cup of water to the bucket gives a digital representation of the rate at which water leaves the bucket.

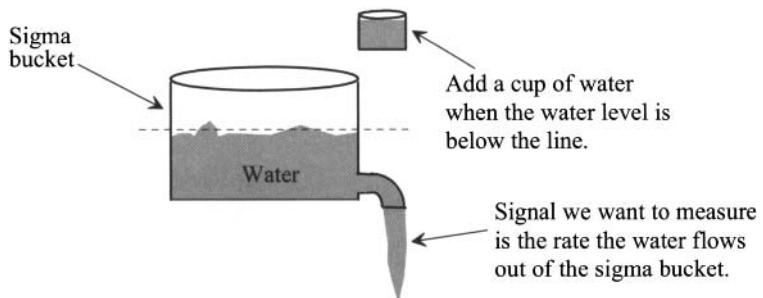


Figure 17.4 Using DSM to measure the water leaving a bucket.

17.1.2 Using DSM for Sensing in Flash Memory

Let's discuss sensing using DSM in a floating gate memory cell technology (Flash). Figure 17.5a shows the characteristics of the Flash memory cell discussed in the last chapter (see Fig. 16.64). When we are sensing the state of the Flash cell, that is, erased or programmed, we hold the row line at ground so that the cell's V_{GS} is 0. If the bit line is held at a potential above ground, say 1 V, then a current of either I_{erased} or I_{prog} (ideally) flows in the cell. In reality variations in the production of the cell and the consistency of the erase from cell to cell affect these values. Using DSM we can more precisely determine the drain current in the cell. This allows us to make a more intelligent decision about the state of the cell. Further, DSM can also be used to program the cell to precisely set the programmed current flow. This allows us to make a memory cell out of a single transistor, which can be used to store several logic levels (values of programmed current). Because of the ability to precisely control the programming operation in a floating gate technology, programming using Fowler-Nordheim Tunneling (FNT) can be abandoned (gate oxides of around 80 Å) for direct tunneling (gate oxides < 20 Å). Of course, the issues with data retention (the charge leaking off the floating gate) are still a concern. In this section we qualitatively discuss how DSM can be used for sensing in Flash technology.

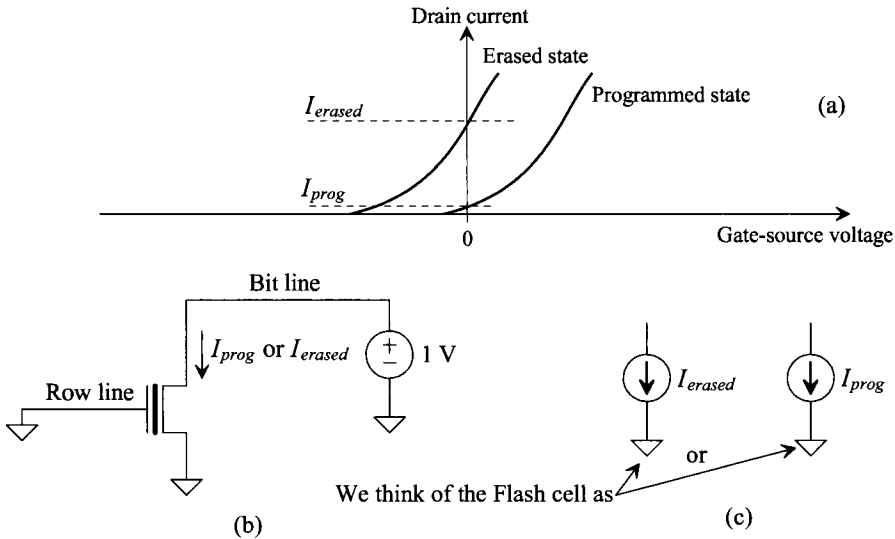


Figure 17.5 The IV characteristics of a NAND memory cell and how we think of the cell when sensing.

The Basic Idea

Figure 17.6 shows the basic idea. The sigma bucket in this figure is the bit line capacitance, C_{bit} . As seen in Fig. 17.4, the signal we are measuring is the rate current flows out of this bucket, that is I_{erased} or I_{prog} . The comparator is used to determine if the voltage on the bit line is below 1 V (again an arbitrary reference). The memory cell is continuously removing charge from the bit line. When the bit line voltage is below 1 V,

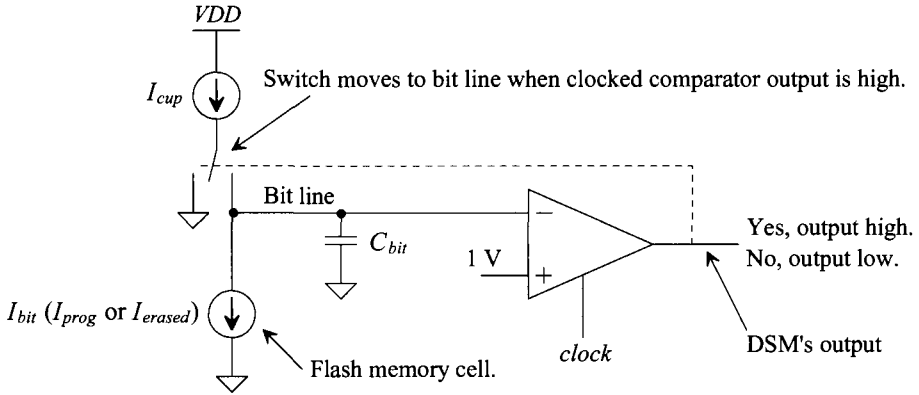


Figure 17.6 Sensing a Flash memory cell using DSM.

the current source, I_{cup} , is connected to C_{bit} to provide the water to the bucket. Note that *we cannot connect the bit line to a voltage source when the output of the comparator goes high*. A voltage source, in our water analogy, is a pressure with an infinite supply of water. The connection of the voltage source would fill the bucket up (the bit line capacitance) to the pressure of the source (the voltage of the source). Again, by looking at the number of times the output of the DSM sensing circuit goes high, we can determine, very precisely, the value of I_{bit} .

If we are clocking the comparator at a rate of f at a period of $T (= 1/f)$, then the rate that charge is removed from the bit line capacitance (the sigma bucket) is

$$\frac{I_{bit}}{C_{bit}} = \frac{\Delta V_{bit}}{T} \quad (17.3)$$

where ΔV_{bit} is the change in the bit line voltage. The amount of charge removed from the bit line in one clock cycle T is

$$Q_{bit} = I_{bit}T = C_{bit} \cdot \Delta V_{bit} \quad (17.4)$$

These quantities are constant. However, the rate at which we add charge to the bit line isn't a constant (unless the output of the comparator is always the same). If N is the total number of clock cycles (the total number of times we clock the comparator) and M is the number of times the output of the comparator goes high (the number of "Yes, add charge to the capacitor" signals), then the rate we add charge to the bit line from I_{cup} is

$$Q_{cup} = I_{cup} \cdot \frac{M}{N} \cdot T \quad (17.5)$$

In order for the voltage on the bit line (the water level in the bucket) to remain, on average, constant, we require that the amount of charge leaving the bucket (Q_{bit}) equal the amount of charge entering the bucket (Q_{cup}) or

$$Q_{bit} = I_{bit}T = Q_{cup} = I_{cup} \cdot \frac{M}{N} \cdot T \quad (17.6)$$

or

$$\frac{I_{bit}}{I_{cup}} = \frac{M}{N} \quad (17.7)$$

This result is important. It relates the counter output code, M , and the total number of times the DSM sensing circuit is clocked, N , to the ratio of the Flash memory cells current, I_{bit} and the fed-back signal I_{cup} . Note again that the charge leaving the bit line capacitance can't be greater than the charge entering the bit line capacitance, that is, we require $I_{cup} \geq I_{bit}$. As mentioned earlier, using a large cup (large I_{cup}) increases the sensing time for a required resolution (we're averaging a larger variable).

Notice that Eq. (17.7) doesn't include the clock frequency or period. We might think that these quantities aren't important. While, if the DSM is designed correctly, it doesn't directly affect the output of the sensing operation, we can have the situation where the bucket (bit line capacitance) empties or overflows (goes to ground or V_{DD}). To avoid this situation, we may require that the maximum deviation on the bit line, $\Delta V_{bit,max}$ be less than some value over a time T . We can write, assuming $I_{cup} \geq I_{bit}$,

$$\Delta V_{bit,max} = \frac{I_{cup}T}{C_{bit}} = \frac{I_{cup}}{C_{bit} \cdot f_{clk}} = I_{cup} \cdot R_{sc} \quad (17.8)$$

For example, if the bit line capacitance is 500 fF, the clock frequency is 100 MHz ($T = 10$ ns), and I_{cup} is 10 μ A, then $\Delta V_{bit,max} = 0.2$ V. If we clock the DSM slower, we must increase our bucket size (add capacitance in parallel with C_{bit} on the input of our DSM).

Example 17.1

Suppose that the DSM sensing circuit in Fig. 17.6 is used to determine the current flowing in a programmed Flash memory cell. If the single transistor Flash memory cell is programmed to conduct 1, 3, 5, or 7 μ A of current, estimate the counter output codes if the DSM is clocked 15 times (a 4-bit counter is used) and I_{cup} is 10 μ A. Estimate the maximum bit line voltage change if the DSM is clocked at 100 MHz and the bit line capacitance is 500 fF.

For the 1 μ A program current, using Eq. (17.7), we get

$$\frac{1}{10} = \frac{M}{15} \rightarrow M = 1.5 \text{ so the counter output would be 2 (0010)}$$

For the 3 μ A program current through the Flash memory cell,

$$\frac{3}{10} = \frac{M}{15} \rightarrow M = 4.5 \text{ so the counter output would be 5 (0101)}$$

For the 5 μ A program current,

$$\frac{5}{10} = \frac{M}{15} \rightarrow M = 7.5 \text{ so the counter output would be 8 (1000)}$$

Finally, for the 7 μ A program current,

$$\frac{7}{10} = \frac{M}{15} \rightarrow M = 10.5 \text{ so the counter output would be 11 (1011)}$$

The maximum deviation of the voltage on the bit line would be when the program current is 1 μ A (leaving the bit line) and I_{cup} is connected (10 μ A charging the bit line). This would give a net current of 9 μ A into the bit line so

$$\Delta V_{bit,max} = \frac{9\mu A \cdot 10\text{ ns}}{500\text{ fF}} = 180\text{ mV}$$

Let's show the output decisions for the case when I_{prog} is $1\text{ }\mu\text{A}$, Table 17.2. Note that if $1\text{ }\mu\text{A}$ is removed from the 500 fF bit line capacitor for 10 ns , the bit line voltage drops 20 mV . If (net) $9\text{ }\mu\text{A}$ of current is put into the bit line for 10 ns , the bit line voltage increases by 180 mV . Note that as calculated, the maximum change in the bit line voltage is 180 mV . Also note that if the comparator had output a Yes at 100 ns instead of at 110 ns , it wouldn't have affected the final output. ■

Table 17.2 See Ex. 17.1.

Time, nanoseconds	Bit line voltage	Add current? ($V_{bit} < 1\text{ V}$?)	Bit line current	Counter output
0	1		$-1\text{ }\mu\text{A}$	0 (0000)
10	0.98	Yes (1)	$9\text{ }\mu\text{A}$	1 (0001)
20	1.16	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
30	1.14	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
40	1.12	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
50	1.1	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
60	1.08	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
70	1.06	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
80	1.04	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
90	1.02	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
100	1	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
110	0.98	Yes (1)	$9\text{ }\mu\text{A}$	2 (0010)
120	1.16	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)
130	1.14	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)
140	1.12	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)
150	1.1	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)

Notice, in the previous example, that we can clock the DSM indefinitely. The only limitation is the size of the counter (after a time the counter will roll over). This is important because, to get better resolution, all we have to do is sense for a longer period of time. We can define the *dynamic range* of the sense operation, in dB, as the maximum counter output code (N) to the smallest output code (1)

$$\text{dynamic range (DR)} = 20 \cdot \log \frac{N}{1} = 20 \cdot \log N \quad (17.9)$$

noting that no signal would give a counter output code of zero. Clocking the DSM 1,000 times results in, ideally, a DR of 60 dB . Clocking the DSM 15 times results in a DR of 23.5 dB .

The maximum input signal, $I_{bit,max}$, equals the fed-back signal I_{cup} ($I_{cup} \geq I_{bit}$). The minimum resolvable signal is then

$$\text{minimum resolvable signal} = \frac{\text{fed-back signal}}{N} \quad (17.10)$$

Again, this illustrates that the larger the fed-back signal, the more averages, N , are needed for a given resolution.

Example 17.2

Determine the minimum resolvable programmed current, I_{bit} , in Ex. 17.1. Verify the answer with a table similar to Table 17.2.

Using Eq. (17.10), the minimum resolvable current is $10 \mu\text{A}/15$ or $0.666 \mu\text{A}$. If this current is removed from the bit line capacitance for 10 ns , the voltage drop is 13.33 mV . If the bit line capacitance is charged with $9.333 \mu\text{A}$ for 10 ns , its voltage increases to 186.66 mV . Table 17.3 tabulates the operation of the DSM sensing circuit when I_{prog} is 666.6 nA . Note that in all situations where I_{prog} is not zero the first decision is a Yes. However, after neglecting this decision, N clock cycles later the counter increments to 2 when the cell is sinking the minimum resolvable signal (here 666.6 nA). ■

Table 17.3 See Ex. 17.2.

Time, nanoseconds	Bit line voltage (millivolts)	Add current? ($V_{bit} < 1 \text{ V}$?)	Bit line current	Counter output
0	1,000		$-0.666 \mu\text{A}$	0 (0000)
10	986.67	Yes (1)	$9.333 \mu\text{A}$	1 (0001)
20	1,173.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
30	1,160	No (0)	$-0.666 \mu\text{A}$	1 (0001)
40	1,146.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
50	1,133.34	No (0)	$-0.666 \mu\text{A}$	1 (0001)
60	1,120	No (0)	$-0.666 \mu\text{A}$	1 (0001)
70	1,106.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
80	1,093.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
90	1,080	No (0)	$-0.666 \mu\text{A}$	1 (0001)
100	1,066.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
110	1,053.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
120	1,040	No (0)	$-0.666 \mu\text{A}$	1 (0001)
130	1,026.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
140	1,013.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
150	1,000	No (0)	$-0.666 \mu\text{A}$	1 (0001)
160	986.67	Yes (1)	$9.333 \mu\text{A}$	2 (0010)

The Feedback Signal

The precision of a sense operation in a DSM is limited by how precisely current can be guided into or out of the bit line capacitance. The bit line capacitance can be thought of as integrating (an integrator, *sigma*) the difference (*delta*) between the fed-back signal (I_{cup} in Fig. 17.6) and the signal we are measuring (I_{bit}). If the amount of charge (current) steered into the bit line isn't consistent from one clock period to the next, errors will occur. To illustrate the possibility of errors, examine the circuit seen in Fig. 17.7. When the switch is connected to ground $C_{parasitic}$ (the parasitic capacitance on the output of the current source), is discharged. Now when the switch is connected to the bit line I_{cup} must charge both $C_{parasitic}$ and C_{bit} . Because the bit line voltage will be moving around and the switch may stay connected to the bitline for two or more consecutive cycles, errors in the sense will occur that limit the resolution. For example, the charge supplied to $C_{parasitic}$ may be $1.05 \cdot C_{parasitic}$ (a bit line voltage of 1.05V) in one clock cycle, while in a different clock cycle it may be $1.02 \cdot C_{parasitic}$. In our water analogy in Fig. 17.4 the amount of water in the cup effectively changes due to this parasitic. If the size of the bit line capacitance is large (or if capacitance is added to the input of the DSM to purposely increase this capacitance) or only a coarse sense is used (say 128 or less clock cycles), then $C_{parasitic}$ will probably not affect the sense in any significant way. Note that if the switch were connected to 1V in Fig. 17.7 instead of ground, the errors from $C_{parasitic}$ would be reduced.

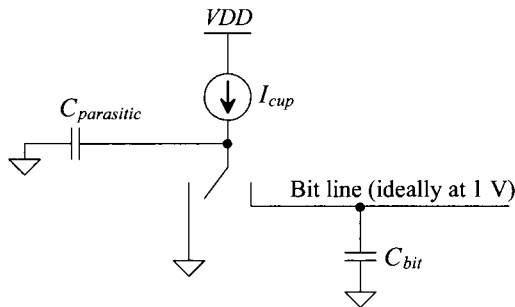


Figure 17.7 How the parasitic capacitance on the output of the current source causes errors.

In an attempt to minimize the power (power is burned unnecessarily when I_{cup} is connected to ground in Figs. 17.6 or 17.7) and the effects of parasitics consider the switched-capacitor (SC) circuit seen in Fig. 17.8. In the following we assume the bit line reference voltage (here 1 V) is greater than the threshold voltage of the PMOS transistors so that the PMOS switches used in the SC can turn fully on. The two clock signals, ϕ_1 and ϕ_2 form nonoverlapping clock signals (they are never low at the same time). This is important because we never want to connect the bit line directly to VDD (which occurs if the clock signals are all low at the same time). When ϕ_1 is low, C_{cup} charges to VDD . Next, ϕ_1 goes high. Between ϕ_1 going high and ϕ_2 going low, the comparator is clocked (on the rising edge of *clock*). If charge needs to be added to the bit line, the comparator output goes high and the gate of M3 is driven low. Note that the inverter can be removed